

POCKET PRODUCER: SESSION-CONDITIONED RETRIEVAL AS USER-CONTROLLED PROPOSAL

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ABSTRACT

We present Pocket Producer, a working prototype that operationalizes *session-conditioned continuation retrieval* inside the Audiotool DAW. A five-signal logistic fusion, trained with BPR on pack co-membership weak labels from the Freesound Loop Dataset, is deployed in the system as a cold-start candidate generator. Four lines of evidence show that this proxy should not be trusted to decide autonomously: (1) the apparent fusion gain is label-sensitive—+4.0pp over frozen cosine on hard_similar negatives when 26% of weak labels group items by uploader rather than pack, but −0.03pp under pack-only labels; (2) leave-one-signal-out ablation reveals regime-dependent signal contributions—tempo helps on hard_similar but hurts on hard_tempo_key; (3) on a held-out split committed to version control before evaluation, the fusion—cosine difference is small and uncertain (hard_similar +1.5pp, 95% CI crossing zero) and significantly negative on hard_tempo_key; (4) an exploratory pilot shows a descriptive reversal between weak-label signal rankings and human consensus. Pocket Producer therefore treats every recommendation as a *proposal*: candidates are previewable, insertion is reversible, and feedback is logged for future personalization—which is not yet evaluated.

1. INTRODUCTION

When a musician returns to an unfinished loop-based project, the question is not “what sounds similar?” but “what would be useful as the next step?” Loop compatibility [1–4], stem coherence [5–7], session-conditioned generation [8], and reversible co-creation [9] all address parts of this problem. We define **continuation utility** as the property that a candidate helps the musician make progress in the current session—unlike similarity, it is asymmetric, context-dependent, and multi-dimensional.

We contribute: (1) a working DAW-integrated prototype whose deployed ranker is the same fusion model evaluated here, treating retrieval as user-controlled proposal, not automatic decision; (2) empirical evidence that weak-label proxy scores are sensitive to label construction, regime-dependent, uncertain under held-out evaluation, and may

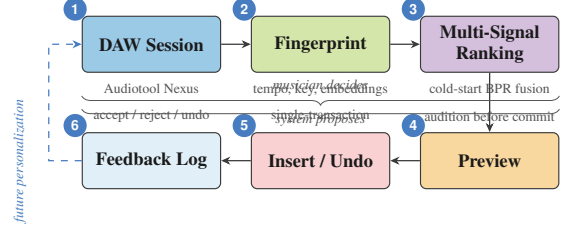


Figure 1. Pocket Producer pipeline: the system proposes (steps 1–3); the musician decides (steps 4–6). Feedback is logged but personalization is not yet trained.

not reflect human judgment—motivating the design choice to keep creative authority with the musician.

2. SYSTEM

Figure 1 shows the pipeline. The live Audiotool session is read via the Nexus SDK and mapped to a session fingerprint (tempo, active regions, CLAP embeddings [10]). A five-signal logistic fusion trained with BPR loss [11] ranks candidates across timbral similarity (CLAP cosine to session regions, mean- and max-pooled), tempo match, key compatibility, and tag overlap. The deployed ranker is this same fusion model, registered as the default whenever session region audio is available, with two stated deltas from the offline evaluation: Audiotool documents carry no key signature, so the key term is constant zero online, and tag context comes from fragments used in the session rather than pack siblings. Without region audio the service falls back to a transparent rules baseline. The system positions itself as a *proposer*: candidates are surfaced for preview—together with per-signal evidence (audio similarity, tempo fit, tag overlap)—before any session state changes. Accepted candidates are inserted via a single Nexus transaction with undo support; rejected or undone events are logged. This design is a response to the empirical findings in Section 4: because the proxy’s correspondence to continuation utility is limited and regime-dependent, the system must not substitute its ranking for the musician’s judgment.

3. WEAK-LABEL FUSION

From the FSLD [12] we kept only sounds with a true Freesound pack (1,797 items, 605 packs), discarding items whose only grouping is a shared uploader. Each multi-item pack yields one leave-one-out query per item,



Table 1. LOSO ablation: accuracy change (pp) when one signal is removed. Pack-only train side, 484 sources, mean over 5 splits. Tempo (bold) shows the clearest regime dependence.

Signal removed	easy	hard_sim.	hard_tk
audio_cos_mean	-0.32	-0.36	-1.04
audio_cos_max	+0.17	-0.30	-0.15
tempo	-0.27	-0.96	+0.68
key	-0.21	+0.94	+1.21
tag_jaccard	+0.09	-0.79	-0.07

with same-pack siblings as positives and cross-pack items as negatives in three difficulty tiers (*easy*: tempo/key mismatch; *hard_tempo_key*: compatible tempo and key; *hard_similar*: highest cosine to the positive from another pack). A 121-source held-out split (seed 20260810) was committed to version control together with the evaluation protocol *before* any evaluation ran; all development below uses only the 484 train sources (981 examples). LAION-CLAP Music [13] (512-dim) provides the embedding backbone.

On the train side (mean over 5 source-grouped splits), frozen cosine reaches 0.929 overall and 0.814 on hard_similar; five-signal fusion reaches 0.925 and 0.813. Under pack-only labels the fusion advantage is gone: -0.03 pp on hard_similar, and on hard_tempo_key fusion is significantly *worse* than cosine (-2.1 pp, paired bootstrap 95% CI [-3.9, -0.6] pp).

4. WHY THE PROXY SHOULD NOT DECIDE ALONE

Four lines of evidence, each bearing a limited conclusion, jointly motivate treating the proxy as a proposal rather than a decision.

4.1 The Gain Is Label-Sensitive

An earlier version of this evaluation used the manifest’s fallback grouping, treating *same-uploader* items as co-members in 26% of examples. On that mixed-label corpus, fusion appeared to beat frozen cosine by +4.0 pp on hard_similar (949 sources, same protocol). Restricting to true pack co-membership erases the gain (Section 3). A proxy whose measured advantage depends this strongly on how weak labels are constructed cannot be treated as a reliable estimate of continuation utility.

4.2 Regime-Dependent Signal Contributions

Table 1 shows leave-one-signal-out deltas per negative regime. Removing tempo costs -0.96 pp on hard_similar (where candidates sound alike and tempo becomes a distinguishing factor) but *gains* +0.68 pp on hard_tempo_key; removing key helps on both hard regimes (+0.94/+1.21 pp). Both patterns replicate directionally on the held-out split (tempo -1.48/+0.37 pp, key +0.62/+0.97 pp, retrained on the train side). Part of this pattern is built in: the easy and hard_tempo_key

regimes are themselves constructed from tempo and key, so a tempo/key signal is near-constant within them by design. Still, a single fused score averages over regimes in which individual signals switch sign, so its ranking cannot be uniformly trusted.

4.3 Held-Out Evaluation

The 121-source held-out split was evaluated exactly once, after all development; protocol and split were committed to version control first, so the freeze is verifiable in the repository history. Fusion was retrained on the 484-source train side with the same hyperparameters and 5 seeds. The fusion-cosine difference on hard_similar is +1.5 pp with a source-grouped bootstrap 95% CI of [-1.5, +4.7] pp (joint resampling shared across seeds)—small and uncertain. On hard_tempo_key, fusion performs significantly *worse* than cosine (-1.0 pp, CI entirely below zero); overall the difference is +0.4 pp with a CI crossing zero.

We note residual limitations: the signal set and hyperparameters were designed on the earlier mixed-label corpus, and the pack-only corpus is a subset of data inspected during that development. This run is a sensitivity re-analysis under a committed split, not de novo confirmatory evidence.

4.4 Exploratory Pilot

Nine external raters labeled 20 blind A/B pairs (Fleiss’ $\kappa = 0.294$; 9 consensus pairs at $\geq 7/9$ majority). On these 9 pairs, harmonic compatibility—a signal with no measurable weak-label contribution in our ablations—matches the human majority in 7/9 cases, while tempo, the strongest hard_similar contributor in Table 1, matches in 3/9—a descriptive reversal of the weak-label ranking. The comparison remains exploratory: the pilot is post-hoc, small, and uses a different corpus and pair construction than the weak-label evaluation. It raises the possibility of misalignment between proxy rankings and human judgment but does not confirm it.

5. CONCLUSION

Pocket Producer demonstrates that session-conditioned retrieval can be operationalized in a DAW while keeping creative judgment with the musician. The deployed weak-label fusion is an *operational* cold-start candidate generator: on the held-out split it performs comparably to frozen cosine overall (+0.4 pp, CI crossing zero), trading a positive but uncertain point estimate on hard_similar (+1.5 pp, 4/5 seeds) against a significant deficit on hard_tempo_key—and, unlike a single opaque cosine score, it surfaces per-signal evidence with each proposal. Because its signal contributions switch sign across regimes and its rankings may not reflect human preference, the proxy never decides alone: every recommendation is previewable and reversible. Feedback logging provides the data foundation for future personalization, which is not yet trained or evaluated.

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AI Usage Statement. Claude Code (Anthropic) was used for engineering implementation, experiment scripting, and documentation drafting. All research claims, experimental design, and analysis are the sole responsibility of the author.

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